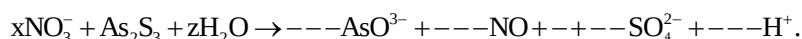


Subjective Assignment (Redox Reaction)

1. When BrO_3^- ion reacts with Br^- ion in acidic medium, Br_2 is liberated. Calculate the ratio of molecular weight and equivalent weight of KBrO_3 .
2. A sample of 28 mL of $\text{H}_2\text{O}_2(\text{aq})$ solution required 10 mL of 0.1 M $\text{KMnO}_4(\text{aq})$ solution for complete reaction in acidic medium. What is the volume strength of H_2O_2 ?
3. In the redox reaction



What is the value of $\frac{x}{2}$?

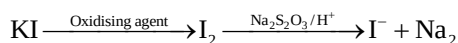
4. On heating 0.220 g of metallic oxide in presence of hydrogen, 0.045 g of water is formed. If the equivalent weight of the metal is E, then what is the value of E/9?
5. 10 g mixture of $\text{K}_2\text{Cr}_2\text{O}_7$ and KMnO_4 was treated with excess of KI in acidic medium. Iodine liberated required 100 cm^3 of 2.2 N sodium thiosulphate solution for titration. IF the mass percent of KMnO_4 in the mixture Z, then what is the value of 2Z/5?
6. In a ore the only oxidizable material is Sn^{2+} . This ore is titrated with a dichromate solution containing 2.5 g $\text{K}_2\text{Cr}_2\text{O}_7$ in 0.50 liter. A 0.40 g of sample of the ore required 10.0 cm^3 of the titrant to reach equivalent point. If the percentage of tin in ore is x, then what is the value of x/17? (K = 39.1, Cr = 52, Sn = 118.7)
7. Read the following data and answer accordingly.

Iodine titrations : Compounds containing iodine are widely used in titrations, commonly known as iodine titration. It is of two kinds :

(i) Iodometric titrations

(ii) Iodometric titrations

(i) Iodometric titrations : It is nothing but an indirect method of estimating the iodine. In this type of titration, an oxidising agent is made to react with excess of KI, in acidic medium or, basic medium in which I^- oxidises into I_2 . Now the liberated I_2 can with $\text{Na}_2\text{S}_2\text{O}_3$ solution.



Although solid I_2 is black and insoluble in water, but it converts into soluble I_3^- ions.



Starch is used as indicator near the end point or equivalence point. Even small amount of I_2 molecules, gives blue colour with starch. The completion of the reaction can be detected when blue colour disappears at the end point. In iodometric titration, the strength of reducing agent is determined by reacting it with I_2 .

Question (i) When 79.75 g of CuSO_4 sample containing inert impurity is reacted with KI, the liberated I_2 is reacted with 50 mL (1M) $\text{Na}_2\text{S}_2\text{O}_3$ in basic medium, where it oxidises into SO_4^{2-} ions, and I_2 reduces into I^- , then what will be the percentage purity of CuSO_4 in sample ?

- (1) 60% (2) 80% (3) 50% (4) 95%

Question (ii) When 214 g of KIO_3 reacts with excess of KI in presence of H^+ , then it produces I_2 . Now I_2 is completely reacted with 1 M $\text{Na}_2\text{S}_2\text{O}_3$ solution in basic medium, where it converts into SO_4^{2-} ions. Then what volume of $\text{Na}_2\text{S}_2\text{O}_3$ is needed to react the end point of the reaction ?

- (1) 500 mL (2) 800 mL (3) 1500 mL (4) 750 mL

Question (iii) A solution containing Cu^{+2} and $\text{C}_2\text{O}_4^{2-}$ ions which on titration with M/10 KMnO_4 requires 50 mL. The resulting solution is neutralized with K_2CO_3 , then treated with excess of KI. The liberated I_2 required 25 mL M/10 $\text{Na}_2\text{S}_2\text{O}_3$ in acidic solution, then what is the difference of the number of m mole of Cu^{+2} and $\text{C}_2\text{O}_4^{2-}$ ions in the solution :

- (1) 40 (2) 10 (3) 30 (4) 50

Question (iv) When 1.66 g of KI is reacted with excess of KIO_2 in presence of 6 N HCl, then ICI is produced. The amount of KIO_3 reacted and the ICI formed are respectively :

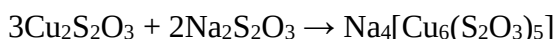
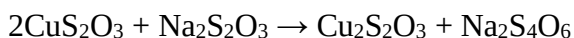
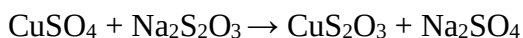
- (1) 4×10^{-2} moles, 3×10^{-3} mole
 (2) 1.5×10^{-2} mole, 5×10^{-3} mole
 (3) 5×10^{-2} mole, 1.5×10^{-3} mole
 (4) 5×10^{-3} mole, 1.5×10^{-2} mole

(v) In order to estimate O_3 in the air, 22.4 L of air at NTP, is passed through acidified KI solution where O_2 is evolved and I is oxidised to I_2 . The liberated I_2 required 200 mL M/10 $\text{Na}_2\text{S}_2\text{O}_3$ in acidic medium, then what will be the % composition of O_3 by volume in the sample ?

- (1) 50% (2) 10% (3) 0.1% (4) 1%

Read the following things and give answers accordingly.

632 g of sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3$) reacts with copper sulphate to form cupric thiosulphate which is reduced by solidum thiosulphate to give cuprous compound which is dissolved in excess of sodium thiosulphate to form a complex compound sodium cuprothiosulphate ($\text{Na}_4[\text{Cu}_6(\text{S}_2\text{O}_3)_5]$).



Sodium cuprothiosulphate

In this process, 0.2 mole of sodium cuprothiosulphate is formed. (O = 16, Na = 23, S = 32)

Now answer the following questions.

Question (i) The average oxidation states of sulphur in $\text{Na}_2\text{S}_2\text{O}_3$ and $\text{Na}_2\text{S}_4\text{O}_6$ are respectively :

- (1) +5 & +2 (2) +2 & +2.5 (3) +5 & 2.5 (4) +2 & +4

Question (ii) Moles of sodium thiosulphate reacted and unreacted after the reaction are respectively:

- (1) 3 & 2 (2) 2 & 3 (3) 2.2 & 1.8 (4) 1.8 & 2.2

Question (iii) If instead of given amount of sodium thiosulphate, 2 moles of sodium thiosulphate along the with 3 moles of CuSO_4 were taken initially. Then moles of sodium cuprothiosulphate formed is :

- (1) 0 (2) 1 (3) 1.5 (4) 2